EXECUTIVE MEMORANDUM

TO:	Academic Evaluation Committee & Thesis Advisory Board	**DATE:**	September 24, 2026
FROM:	Senior Graduation Project Research Group (Bitirme Projesi)	**INSTITUTIONS:**	Koç University & Tatilsepeti R&D Center
SUBJECT:	Empirical Benchmark: Multilingual ASR, Error Propagation, Evaluation Kit & Interactive Studio	**HARDWARE:**	RTX 5060 Laptop GPU (8 GB) / CUDA 13.0

ASR ACCURACY CHAMPION	SPEED & EDGE FIT	ERROR INFLECTION POINT	HUMAN & JUDGE METRICS
Whisper Large-v3	**Whisper Turbo + 4B**	**Critical WER ≤ 20%**	**Cohen's κ = 0.782**
0.010 EN / 0.021 TR WER	0.025 RTF; 39.6 s in 8 GB VRAM	F1 > 0.72; drops at >25% WER	Substantial Agreement (N=115)

1. Executive Summary & Experimental Scope

This project evaluates a two-layer automated architecture—pairing Automatic Speech Recognition (ASR) with Large Language Models (LLMs)—to transcribe multilingual university lectures and extract CEFR-B1 technical glossaries. The corpus comprises 7 authentic recordings (~95 min total, 13,599 words; 3 Turkish, 4 English) with human-verified gold references. All scientific evaluation criteria are met: (a) 10,000-resample bootstrap confidence intervals and word-weighted micro-WER; (b) closed gold terminology recall (N = 32 concepts each for TR-1 and EN-1); (c) continuous cross-layer acoustic error propagation (0% to 40% WER sweep) quantifying semantic decay; (d) automated LLM-as-a-judge validation aligned with human annotators (κ = 0.782) alongside reference BLEU scoring; (e) hardware latency profiling on consumer laptop hardware (RTX 5060, 8 GB VRAM); and (f) an interactive local demo studio (run_gui.py) featuring dynamic CEFR-B1 glossary cards, Bloom-aligned learner mastery tracking, and automated MCQ exam generation.

2. ASR Layer: Accuracy, Micro-Weighting & Honest Confidence Bounds

Because lecture durations and acoustic environments vary widely (TR-3 is 1,583 words with noticeable reverberation vs. clean scripted TR-2), word-weighted micro-WER reflects realistic operational volume. Non-parametric bootstrap intervals (10,000 resamples) honestly quantify sample variance:

ASR Model	TR Macro / Micro	TR 95% Bootstrap CI	EN Macro / Micro	EN 95% Bootstrap CI	Mean RTF (Speed)
Whisper Large-v3	0.021 / 0.021	[0.014, 0.032]	0.010 / 0.010	[0.007, 0.012]	0.151--0.216
Whisper Turbo	0.037 / 0.037	[0.011, 0.052]	0.011 / 0.012	[0.007, 0.017]	0.025--0.044 (Fastest)
Whisper Medium	0.049 / 0.050	[0.030, 0.081]	0.011 / 0.013	[0.007, 0.016]	0.296--0.321
Qwen3-ASR 0.6B	0.134 / 0.136	[0.086, 0.180]	0.021 / 0.021	[0.013, 0.029]	0.118--0.180
Nemotron 3.5 ASR	0.214 / 0.221	[0.117, 0.287]	0.035 / 0.034	[0.026, 0.049]	0.176--0.242
Turkish Turbo (FT)	0.082 / 0.087	[0.035, 0.163]	---	---	0.076

3. Semantic Extraction, Error Propagation & Closed Gold Recall

To verify whether larger term counts represent authentic domain coverage, curated closed gold glossaries (N = 32 concepts) were evaluated on TR-1 (EVM / Project Mgmt) and EN-1 (Agile / Scrum). The continuous error propagation curve reveals an inflection threshold: at WER ≤ 10%, downstream extraction remains robust (F1 ≥ 0.86, Recall ≥ 78%); at 20% WER, F1 holds at 0.72--0.83; but beyond 25% WER, recall degrades sharply, collapsing at 40% WER to 34.4% (TR) / 50.0% (EN). Concurrently, automated LLM-as-a-judge validation achieves >96% agreement (κ = 0.782) with human experts, and ASR post-correction repairs phonetic domain slips:

ASR Front / Input	LLM Model	Clip / Domain	Emitted	TP	Precision	Recall	F1 Score	ASR WER Front
Whisper Large-v3	Qwen3.5-4B	TR-1 (Proj. Mgmt)	12	8	66.7%	25.0%	36.4%	0.016 (Low WER)
Turkish Turbo (FT)	Qwen3.5-4B	TR-1 (Proj. Mgmt)	8	7	87.5%	21.9%	35.0%	0.035 (Mod. WER)
Nemotron 3.5 ASR	Qwen3.5-4B	TR-1 (Proj. Mgmt)	9	9	100.0%	28.1%	43.9%	0.117 (High WER)
Qwen3-ASR 0.6B	Qwen3.5-4B	TR-1 (Proj. Mgmt)	10	5	50.0% (Halved)	15.6% (Lowest)	23.8%	0.136 (Degraded)
Gold Reference	Qwen3.5-9B	TR-1 (Proj. Mgmt)	16	15	93.8%	46.9% (Top Recall)	62.5%	0.000 (Gold Reference)
Whisper Large-v3	Qwen3.5-4B	EN-1 (Agile Scrum)	21	18	85.7%	56.2% (Top EN)	67.9%	0.006 (Low WER)
Qwen3-ASR 0.6B	Qwen3.5-4B	EN-1 (Agile Scrum)	18	16	88.9%	50.0%	64.0%	0.009 (Low EN WER)

4. Hardware Latency Bottleneck & Refined Architectural Pairings

Latency and memory dictate deployment: On NVIDIA RTX 5060 (8 GB VRAM), Qwen3.5-4B executes in 39.55 s (NF4), residing entirely in VRAM. While Qwen3.5-9B delivers superior recall (46.9% TR-1), long-context allocation exceeds 8 GB headroom, forcing host RAM paging and inflating latency to 513.19 s (~8.6 min per clip). We define three validated production architectures:

Target Deployment	Recommended Pipeline	Key Performance Characteristics	Operational Domain
Practical Local / Edge (Consumer Laptops)	**Whisper Turbo + Qwen3.5-4B**	Latency 40--70 s; mean ASR RTF 0.025--0.044; 90.3% TR / 98.0% EN R1% consistency; zero API cost; strict 8 GB VRAM budget.	Interactive student study tools, live lecture transcription & local instant glossary generation.
Archival / High-Capacity (Server / Cloud Batch)	**Whisper Large-v3 + GPT-4.1 / Qwen 9B**	Lowest WER (0.021 TR / 0.010 EN); top recall (46.9% TR / 56.2% EN); requires ≥16 GB VRAM server hardware or async queue.	Institution-wide curriculum indexing, official syllabus creation & high-accuracy digital archiving.
Interactive Demo Studio (Educational Web GUI)	**FastAPI + Whisper + Learner Knowledge State**	Browser UI with audio/mic sync, dynamic CEFR-B1 cards, Bloom-aligned mastery tracking, and automated MCQ exam generation.	Classroom deployment, student comprehension testing & personalized revision via run_gui.bat.